

Yancheng Liu

Staff Data Engineer

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Profile

Staff Data Engineer with 8+ years of experience designing and delivering end-to-end data solutions for real-time analytics, business intelligence, and AI applications. Strong background in data pipelines, modeling, and platform architecture across AWS, Azure, and GCP environments. Delivered high-impact data solutions in e-commerce, logistics, and content, enabling faster insights, reliable reporting, and smarter decisions through strong cross-team collaboration.

Skills

Language: Python, SQL, PySpark, Scala, Bash

Pipeline & Orchestration: Apache Airflow, dbt, Apache Spark, Delta Lake, Kafka

Warehousing & Lakehouse: Snowflake, Databricks, Redshift, BigQuery

Cloud Ecosystems: AWS, Azure, GCP

BI & Visualization: Looker, Tableau, Power BI, Metabase

AI & ML Pipelines: MLflow, HuggingFace Transformers, Databricks ML

Graph & NoSQL Technologies: Neo4j, AWS Neptune, DynamoDB, ClickHouse

Versioning & Testing: Github, dbt Cloud, CI/CD, dbt tests, observability tools

Education

Master of Science - MS, Computer Science, Georgia Institute of Technology
2013 – 2015

Professional Experience

Airbnb, Staff Data Engineer

02/2022 – Present

- Built the Host Profile Data Service for Airbnb Rooms using Scala, Kafka, and Flink, integrating host activity into Snowflake pipelines that powered the Host Passport feature and improved data freshness by about 23%.
- Designed the Atlas Schema and Databricks + Spark pipelines for Airbnb Services & Experiences data into a unified model that reduced query latency by nearly 20% and simplified business reporting.
- Created Looker dashboards on top of dbt and Snowflake, providing actionable engagement and revenue metrics for product and business teams.
- Developed ML feature pipelines with Python, TensorFlow, and Vertex AI, enhancing personalization and boosting trip-planning engagement.
- Automated 200+ Airflow workflows in AWS with Terraform, maintaining 99% reliability and full observability through Prometheus and Grafana.
- Tuned Delta Lake performance on Databricks, reducing refresh times and compute costs through incremental processing and optimized partitioning.
- Collaborated with PMs, Data Scientists, and UX teams to align data models with key KPIs, ensuring every release delivered measurable business impact.

Meta

Senior Data Engineer

02/2019 – 12/2021

- Rebuilt a large-scale A/B testing pipeline using Spark and Snowflake, improving data freshness by 16% and reliability across thousands of experiments.
- Modernized the Metric Monitor with new streaming and batch pipelines, reducing alert latency by 17% and improving anomaly detection accuracy.
- Optimized Spark + Presto jobs experimentation analysis, improving compute efficiency and reducing job failures through query tuning and parallel execution.
- Built Airflow + Snowflake pipelines for experiment metrics, automating ingestion and reporting to serve product and analytics teams.
- Partnered with Product Science and Infrastructure teams to standardize experiment schemas and develop interactive dashboards in Scuba, improving data visibility and decision speed.

Senior Data Engineer

09/2017 – 02/2019

- Built a new client-side logging pipeline for Instagram Stories, capturing high-volume engagement with improved throughput and data consistency.
- Designed and deployed an end-to-end data solution for Explore Brand Safety, integrating content integrity models with ad-delivery metrics for safer recommendations.
- Developed exposure-based A/B testing metrics for Instagram Ads, enabling fine-grained performance attribution and automated experiment analysis.
- Collaborated with cross-functional ML and Ads Engineering teams to align event schemas and experiment frameworks across Instagram and Facebook.
- Drove adoption of modern data modeling and monitoring standards, improving visibility and reliability of metrics powering Instagram's ad-ranking systems.

Athenahealth, Senior MTS/Developer, Analytics Platform

09/2016 – 08/2017

- Built a large-scale data ingestion system to transport distributed relational data into a cloud-based MPP warehouse, driving over 10x reduction in query latency for analytics and reporting workloads.
- Developed a HIPAA-compliant transformation service that enabled secure, self-service data access for client analytics while maintaining regulatory integrity.